Internet Crimes and Cyber Security

Week: 06 **Session: 28**

Topic : Understanding Threats from Dark Web

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Understanding Threats from Deep and Dark Web

3. Tools for Accessing Dark web

Tor Browser

Tor Browser remains the gold standard for dark web access in 2025. It routes your traffic through multiple encrypted nodes using onion routing technology, effectively masking your IP address and digital footprint.

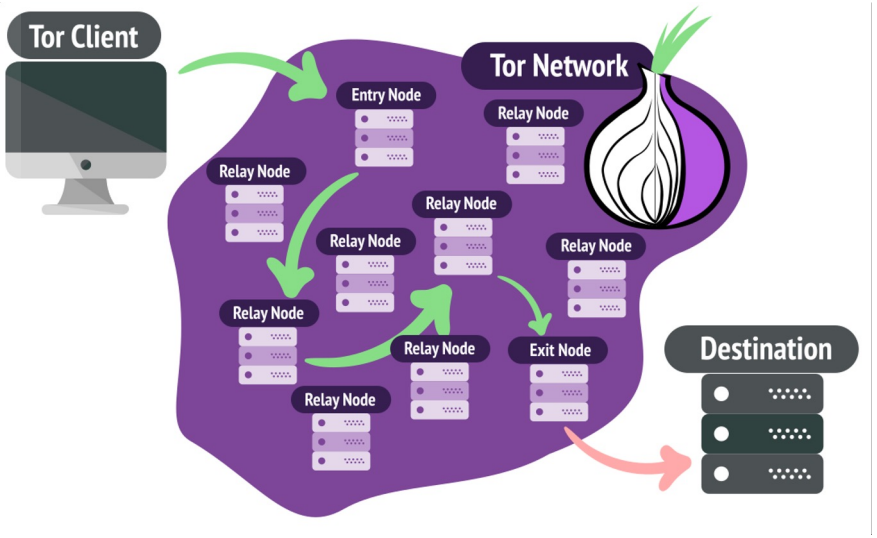
Brave Browser integrates Tor technology directly into its platform, offering users an accessible, user-friendly gateway to the dark web.

Freenet

Freenet operates uniquely, functioning as a decentralized, peer-to-peer network rather than a traditional browser. It encrypts and distributes data across a vast, anonymous network of user nodes, ensuring extreme levels of anonymity and censorship resistance.

I2P Browser

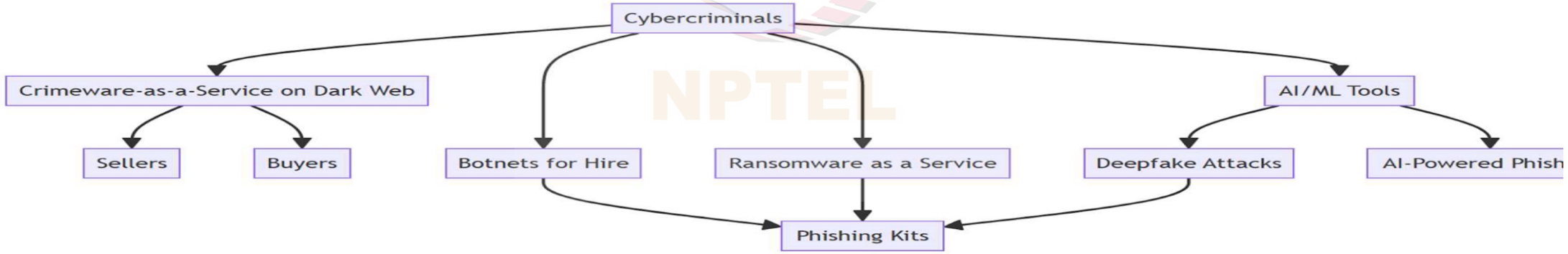
The Invisible Internet Project (I2P) Browser emphasizes a fully encrypted, decentralized approach to dark web access, routing traffic through an anonymous network of user-operated tunnels. It supports both web browsing and secure communication, offering additional services like email and instant messaging fully integrated into its secure environment.



Understanding Threats from Deep and Dark Web

4. Identify cybercrime threats emerging from Dark Web

Emerging Dark Web threats center on the professionalization of cybercrime, offering **Cybercrime-as-a-Service** **(CaaS)** like ransomware kits, stolen data (credentials, corporate access), malware, zero-day exploits, and advanced tools for identity theft and fraud, fueled by cryptocurrencies and facilitated by anonymity, making attacks scalable, targeted, and harder to trace. Threats include selling corporate network access, botnet rentals, custom phishing kits, and advanced persistent threats (APTs), alongside traditional illegal goods,



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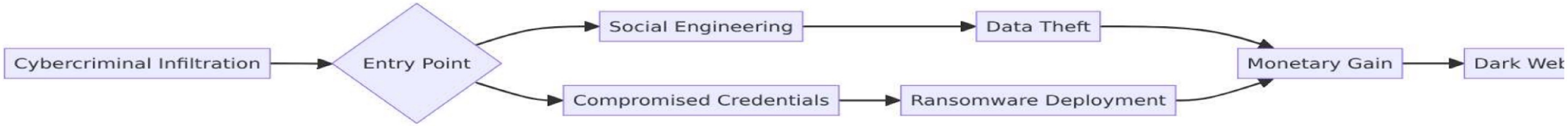

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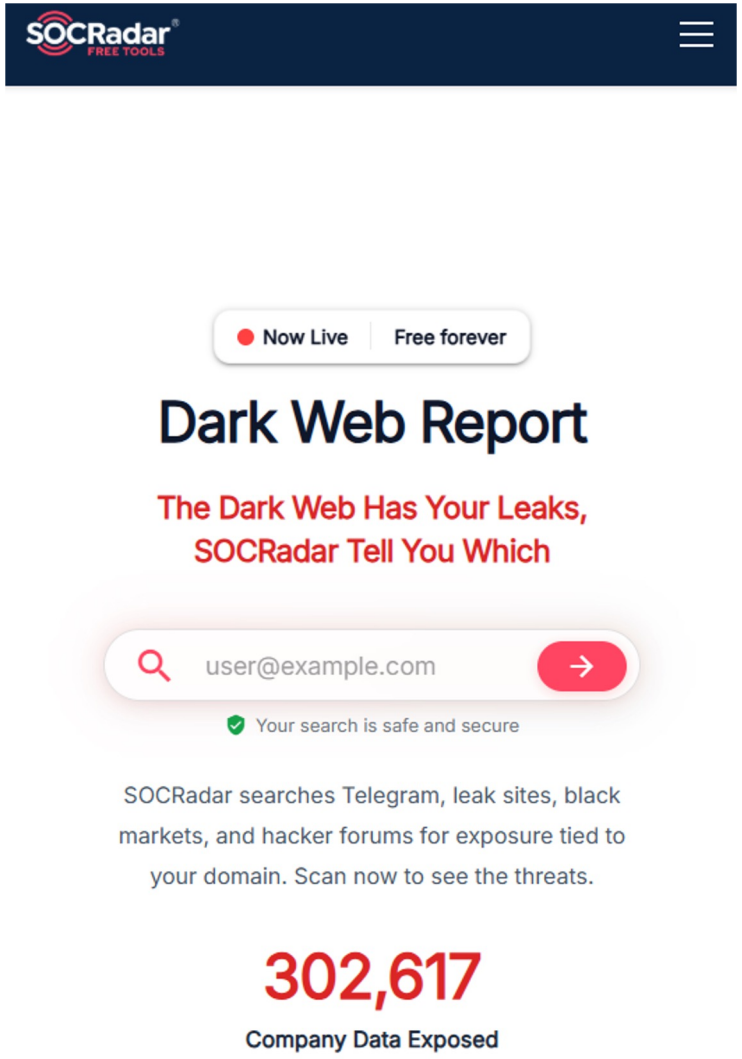
Understanding Threats from Deep and Dark Web

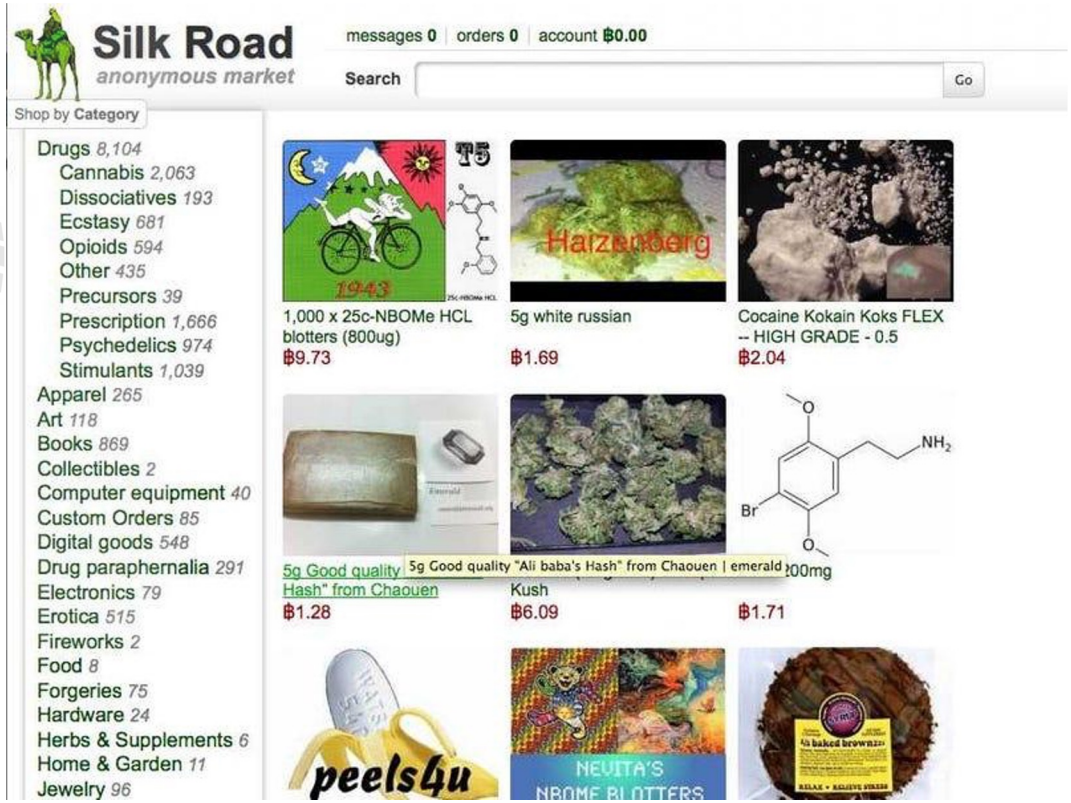
5. Understanding investigative challenges in Dark Web

- **Anonymity & Encryption:** Tools like Tor route traffic through multiple nodes, obscuring IP addresses and making source tracing difficult, especially with cryptocurrency use.
- **Jurisdictional Issues:** Crimes often span multiple countries, complicating legal processes and investigations for any single authority.
- **Evidence Fragility:** Digital evidence is easily altered, deleted, or corrupted, requiring strict chain-of-custody for court admissibility.
- **Technical Skill Gap:** Law enforcement often lacks officers trained in advanced encryption, blockchain, and specific dark web tools.
- **Data Volatility & Searchability:** Dark Web sites (hidden services) have ephemeral addresses, making them hard to find and track compared to the indexed surface web.
- **Malware & Security Risks:** Investigators face constant threats from malware and hacking attempts, requiring secure, isolated environments.
- **Legal & Ethical Dilemmas:** Accessing illicit content poses risks and requires careful navigation to avoid breaking laws or encountering disturbing material.



6. Analyze real-world cases of Cyber Crimes in Dark Web

Silk Road: The most famous early dark web marketplace, Silk Road operated as a Tor hidden service and facilitated the anonymous global trade of illegal drugs and other contraband using Bitcoin. Its founder, Ross Ulbricht (known as "Dread Pirate Roberts"), was eventually tracked down by the FBI and sentenced to double life imprisonment. The shutdown of Silk Road led to the rise and subsequent closure of other markets like AlphaBay and Hansa Market by law enforcement agencies.



Thank You